SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

ONLINE INTERNSHIP & PLACEMENT PREPARATION PORTAL

Team 4 | Document Version: 1.0 (Baseline Specification)
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ACADEMIC INTEGRITY AND PROJECT STATUS DECLARATION

This Software Requirements Specification document describes the requirements, system architecture, interfaces, and verification criteria for the Online Internship & Placement Preparation Portal. As of Document Version 1.0, the project is strictly in Phase 1 (Requirements Phase). No production implementation claims or external dependencies are asserted.

Document Control & Approvals

Revision History

Version	Date	Author(s)	Change Summary	Approval
1.0	05-09-2026	Team 4: Dhanush Rao, Gunavathi M, Darshan R Mali, Deepika K	Initial baseline SRS document covering 24 Functional Requirements, 8 NFRs, 6 Security Requirements, Acceptance Tests, UML Models, and RTM.	Under Review

Project Approvals

Role	Name	Designation / Email	Date
Project Faculty Guide	Dr. / Prof. Project Guide	Faculty Advisor, Dept. of CSE	Pending Review
Course Coordinator	Software Engineering Faculty	Course Lead, Dept. of CSE	Pending Review

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document provides a complete, formal, and unambiguous definition of the software requirements for the Online Internship & Placement Preparation Portal. Developed by Team 4 as part of the Software Engineering curriculum, this specification details functional and non-functional requirements, external interfaces, quality attributes, system models, and verification criteria. The primary objective is to establish a rigorous, verified baseline that governs subsequent engineering phases—specifically architectural design, database modeling, frontend and backend implementation, automated testing, and academic evaluation.

1.2 Scope & System Boundaries

The Online Internship & Placement Preparation Portal is a unified web-based platform designed to consolidate two traditionally disparate academic placement workflows: (1) discovering and applying for internship opportunities posted by registered companies and administrators, and (2) preparing for competitive campus recruitment drives through timed aptitude assessments and progress analytics.

The operational scope of the portal is strictly defined across three primary actors:

- • Student Capabilities: Account creation, academic profile and skills management, PDF resume upload (<= 5 MB), browsing and multi-criteria searching of approved internships, automated eligibility verification, one-click application submission with duplicate prevention, real-time application status tracking, timed aptitude test execution (Quantitative Aptitude, Logical Reasoning, Verbal Ability), instant test scoring and solution review, test history archiving, performance analytics, and receipt of in-portal notifications.
- • Company Capabilities: Native portal registration, organizational profile management, submission of profile details for administrative verification, creation and editing of internship listings, deactivation/closure of listings, viewing received applications, reviewing applicant profiles and accessing authorized PDF resumes, shortlisting or rejecting applicants, and updating candidate application status.
- • Administrator Capabilities: Secure login, verification and approval/rejection of company registrations, activation/deactivation of user accounts, direct creation and moderation of internship listings, curation of the aptitude-test question bank, configuration and publication of timed aptitude tests, oversight of application volumes, and viewing portal-wide analytics.

STRICT REALISTIC PROJECT SCOPE BOUNDARIES

1. No External Job Scraping: The portal does NOT scrape LinkedIn, Indeed, Internshala, or any external website. All listings originate exclusively from registered, approved companies or system administrators.
2. No Simulated/Fictitious Company Partnerships: The system makes no claim of pre-integrated external corporate APIs. Companies must register natively through the portal.
3. In-Portal Notifications: Alerts (such as application status transitions and test announcements) are delivered entirely within the portal UI. External SMS gateways and SMTP email delivery infrastructure are explicitly excluded from the baseline college project scope.
4. No Complex AI/ML or Microservices: Scoring, filtering, and analytics are computed deterministically using standard Node.js algorithms and MongoDB queries. Python microservices and complex distributed architectures are not part of the baseline project.

1.3 Intended Audience & Reading Suggestions

This specification is intended for the following stakeholders:

- • Faculty Evaluators and Project Guides: Review Section 1, Section 4 (Functional Requirements), Section 6 (Acceptance Tests), Section 7 (UML Models), and Section 8 (RTM) to assess project completeness, architectural soundness, and alignment with software engineering methodologies.
- • Team 4 Developers: Consult Section 3 (Interfaces), Section 4 (Functional Specifications), Section 5 (NFRs & Security), and Section 7.3 (Conceptual Domain Relationship) as the foundational contract during Phase 2 (Design) and Phase 3 (Implementation).
- • QA & Test Engineers: Utilize Section 4 (Acceptance Criteria), Section 6 (Acceptance Tests), and Section 8 (RTM) to design unit, integration, and end-to-end verification suites.

1.4 Definitions, Acronyms, and Abbreviations

Table 1.1 provides formal definitions for acronyms and specialized terms utilized throughout this document.

Term / Acronym	Formal Definition and Context
SRS	Software Requirements Specification; the formal document specifying software capabilities and constraints.
OIP	Online Internship & Placement Portal; the standardized prefix for project requirement identifiers.
MERN Stack	The targeted technology stack: MongoDB (database), Express.js (backend web framework), React.js (frontend user interface), and Node.js (JavaScript runtime environment).
JWT	JSON Web Token (RFC 7519); an open standard for securely transmitting digitally signed user identity claims.
RBAC	Role-Based Access Control; security mechanism that restricts API access based on authorized user roles (Student, Company, Admin).
REST	Representational State Transfer; architectural style for networked web applications utilizing standard HTTP methods.
API	Application Programming Interface; formal contracts enabling client-server communication via JSON.
RTM	Requirements Traceability Matrix; grid linking functional requirements to design specifications and test cases.
NFR	Non-Functional Requirement; quality attribute or operational constraint governing performance, reliability, or usability.
FR	Functional Requirement; statement specifying an explicit behavioral capability the software shall execute.
WCAG	Web Content Accessibility Guidelines; W3C standards governing digital accessibility compliance.

1.5 Document References and Standards

- **1.** IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications.
- **2.** ISO/IEC/IEEE 29148:2018: Systems and software engineering — Life cycle processes — Requirements engineering.

- **3.** W3C Web Content Accessibility Guidelines (WCAG) 2.1: Formal guidelines for digital accessibility.

2. Overall Description

2.1 Product Perspective

The Online Internship & Placement Preparation Portal is a self-contained, 3-tier web application engineered to centralize campus placement activities. It eliminates the need for students to navigate disconnected tools (e.g., spreadsheet trackers, physical noticeboards, third-party mock test portals) by providing an integrated digital ecosystem.

The application architecture consists of three cohesive tiers:
1. Presentation Tier (Client): Built using React.js, delivering a single-page application (SPA) with responsive dashboards for Students, Companies, and Administrators.
2. Application Tier (Server): Powered by Node.js and Express.js, exposing stateless RESTful endpoints for business logic, eligibility evaluations, test timing, score calculations, and role-based route protection.
3. Persistence Tier (Database): Implemented using MongoDB, utilizing document collections to model users, profiles, internships, applications, questions, test attempts, and in-portal notifications.

2.2 Major Product Functions

The system delivers nine core functional capabilities:

- **1.** Role-Based Authentication: Secure registration, password hashing via bcrypt, and stateless session management using signed JWTs for Students, Companies, and Admins.
- **2.** Student Profile & Resume Repository: Comprehensive tracking of student academic details (CGPA, degree, branch, year), technical skills, and secure upload/validation of PDF resumes (<= 5 MB).
- **3.** Company Management & Vetting: Corporate profile management, recruiter credentials, and administrative approval workflow before internship publication permissions are granted.
- **4.** Internship Lifecycle Management: Creation, modification, and deactivation of internship postings by approved companies and administrators.
- **5.** Search, Discovery & Filtering: Multi-attribute search allowing students to filter listings by keyword, skill tag, location, work arrangement (On-site/Remote), and stipend.
- **6.** Application Processing & Duplicate Prevention: Strict eligibility checks (CGPA and skill matching), one-click application submission, duplicate application prevention, and applicant status tracking across five distinct states (Applied, Under Review, Shortlisted, Rejected, Selected).
- **7.** Aptitude Question Bank Curation: Admin-governed question repository spanning Quantitative Aptitude, Logical Reasoning, and Verbal Ability, with difficulty classifications and detailed solution explanations.
- **8.** Timed Test Engine & Analytics: Countdown-timer-enforced testing with automatic submission upon timeout, immediate score calculation, test history records, and preparation analytics identifying strong and weak topics.
- **9.** In-Portal Notifications: In-portal notifications displayed through the notification center for application-status changes and newly published aptitude tests.

2.3 User Roles and Characteristics

The system defines three mutually exclusive actors:

2.3.1 Student

Students are university undergraduates or postgraduates preparing for campus placements and seeking internships. Characteristics include moderate technical literacy, expectation of rapid response times (< 2 seconds), reliance on responsive desktop and mobile viewports, and the need for clear visibility into eligibility criteria, deadlines, and preparation progress.

2.3.2 Company

Company actors are corporate recruiters and HR personnel managing internship recruitment. Characteristics include professional workflow expectations, reliance on desktop browsers, need for efficient candidate filtering, authority to inspect applicant resumes, and responsibility for updating application statuses.

2.3.3 Administrator

Administrators are placement coordinators with elevated system privileges. Characteristics include authority to approve or reject company registrations, moderate internship listings, curate the question bank, configure aptitude tests, and monitor overall portal activity.

2.4 Operating Environment

The portal operates within standard modern web environments:

- • Client Environment: Any modern web browser supporting ECMAScript 6+ and HTML5 (Google Chrome v110+, Mozilla Firefox v110+, Apple Safari v16+, Microsoft Edge v110+) on Windows, macOS, Linux, Android, or iOS.
- • Server Environment: Node.js runtime (LTS v20.x or higher) running Express.js on an Ubuntu Linux 22.04 LTS or standard development environment.
- • Database Environment: MongoDB Community Server (v7.0+) accessed via Mongoose ODM.
- • Network Environment: Standard HTTP/HTTPS network communication over standard ports (80/443).

2.5 Design and Implementation Constraints

1. Architectural Decoupling: The backend shall expose stateless REST APIs adhering to JSON specifications, completely decoupled from the React.js client.
2. Stack Uniformity: Implementation is strictly constrained to the JavaScript / MERN Stack (MongoDB, Express.js, React.js, Node.js). No separate Python microservice is required for baseline operations.
3. File Storage Constraints: Resume uploads are strictly constrained to PDF format (.pdf) with an enforced file size ceiling of 5 MB to prevent server resource exhaustion.
4. Data Isolation & Authorization: Role-based access control and company-level authorization shall ensure that students can access only their own profiles, applications, and test history, while companies can view only applicants associated with their own internship postings.
5. Standards Compliance: Strict adherence to WCAG 2.1 Level AA accessibility standards, responsive mobile-first UI design, and industry-standard password hashing using bcrypt.

2.6 Assumptions and Dependencies

1. Network Connectivity: Users possess standard internet connectivity of at least 512 kbps bandwidth.
2. Modern Browser Support: Users access the portal using modern browsers with JavaScript enabled.
3. Truthful Academic Information: Students enter accurate CGPA and academic details during profile registration.
4. Internal Persistence: All internship data, applications, and aptitude tests originate and reside within the portal's MongoDB database.

3. External Interface Requirements

3.1 User Interfaces

The user interface shall be delivered entirely via web browsers, employing a clean, accessible, and responsive dashboard design conforming to WCAG 2.1 AA standards. Specific interface modules include:

- • Student Dashboard: Overview displaying application status cards (Applied, Under Review, Shortlisted, Rejected, Selected), quick search, recently added listings, aptitude readiness metrics, and an in-portal notification bell icon with an unread badge counter.
- • Company Dashboard: Metric counters for Active Listings, Total Applicants Received, and Shortlisted Candidates; tabular listing management with edit/deactivate controls; and candidate review modal for accessing authorized PDF resumes.
- • Admin Console: System-wide metrics (Registered Students, Verified Companies, Pending Approvals, Total Applications); approval management tables; question bank editor; test configuration forms; and portal activity audit view.
- • Aptitude Test Interface: Clean, distraction-free testing layout featuring a persistent countdown timer, question navigation palette showing answered/unanswered states, single-question viewport with radio options, and an explicit submit confirmation dialog.

3.2 Hardware Interfaces

As a web-based software application, the system has no dedicated hardware interfaces. It operates over standard computing hardware:

- • Client Hardware: Any x86_64 or ARM-based desktop, laptop, tablet, or smartphone equipped with minimum 2 GB RAM and a display resolution of at least 360x640 pixels (mobile) or 1280x720 pixels (desktop).
- • Server Hardware: Standard development or virtual private server with minimum 2 vCPUs, 2 GB RAM, and 15 GB persistent storage.

3.3 Software Interfaces

The portal interfaces with the following software packages and runtimes:

- • Operating System: Linux (Ubuntu 22.04 LTS) or cross-platform operating systems for local development and hosting.
- • Node.js Runtime & Express.js: Node.js (v20 LTS) running Express.js for REST routing and middleware execution.
- • Database Management System: MongoDB (v7.0+) managed through Mongoose ODM for schema validation and indexing.
- • File Upload Engine: Multer middleware for handling multipart/form-data resume uploads with MIME inspection.

3.4 Communications Interfaces

All client-server communications shall strictly utilize standard web protocols:
1. Transport Layer Security (TLS): All HTTP traffic shall be encrypted using TLS 1.2+ over TCP port 443 (HTTPS).
2. RESTful JSON Payloads: All API requests and responses shall utilize UTF-8 encoded JSON (RFC 8259) with explicit Content-Type: application/json headers.

3. In-Portal Notification Communication: Standard client-server communication over HTTPS to fetch in-portal notifications and unread badge counts.

4. System Features (Detailed Functional Requirements)

This section details the functional capabilities of the Online Internship & Placement Preparation Portal across its eleven major operational modules. Following the module descriptions, a comprehensive requirements specification table defines exactly 24 testable, unambiguous functional requirements conforming to the mandatory 'shall' convention.

4.1 User Authentication & Session Management

Provides role-based registration, credential validation, secure authentication, and session control. The module enforces strong password policies, hashes secrets using bcrypt with a work factor of at least 10, and issues signed JSON Web Tokens (JWT) containing cryptographically signed user IDs and role claims (Student, Company, Admin). It supports explicit logout with client-side token disposal.

4.2 Student Profile & Resume Management

Enables students to maintain their academic resume within the portal. Captured attributes include full name, institutional roll number, degree, branch, graduation year, CGPA (validated between 0.00 and 10.00), and technical skills. It provides secure upload, replacement, and validation for PDF resumes strictly limited to 5 MB.

4.3 Company Profile Management & Verification

Allows company recruiters to register and build an organizational profile, including company name, website domain, industry sector, headquarters location, and recruiter contact details. Newly registered company accounts are placed in an initial 'Pending Approval' state until verified by the administrator.

4.4 Internship Management & Posting Lifecycle

Empowers verified companies and administrators to create, edit, and deactivate internship listings. Each posting defines a role title, description, required technical skills, minimum CGPA threshold, eligible graduation batches, work arrangement (On-site, Remote, Hybrid), location, duration, monthly stipend, application deadline, and available openings.

4.5 Internship Search, Filter & Discovery Engine

Delivers an intuitive search experience for students. Supports free-text keyword matching across titles, companies, and skill tags, combined with multi-parameter filters (minimum stipend, location, work arrangement, and eligibility match). Results are paginated with sorting options.

4.6 Internship Application Management & Tracking

Governs the end-to-end application lifecycle. When a student applies, the system verifies that the student satisfies all specified eligibility criteria (CGPA, required skills) and has uploaded a resume. It strictly enforces that a student may submit at most one application per unique internship listing. The application record establishes the core conceptual relationship connecting Student, Internship, and Company. Companies can review applicants, inspect resumes, and transition status across Applied, Under Review, Shortlisted, Rejected, and Selected.

4.7 Aptitude Test Management & Question Bank Curation

Furnishes administrators with a comprehensive assessment management console. Questions are cataloged across three primary placement evaluation categories: Quantitative Aptitude, Logical Reasoning, and Verbal Ability. Each question record contains the question stem, four multiple-choice options, correct option index, difficulty level, topic tag, and an explanatory solution.

4.8 Test Engine, Countdown Timer & Automatic Scoring

Provides a timed testing environment for students. Tests can be configured by administrators with specific categories, duration, and question sets. The client runs a synchronized countdown clock; if the time limit expires, the system automatically submits the test state. The server executes instant grading, updates student assessment records, and displays the score breakdown alongside detailed solutions.

4.9 Progress Tracking, Performance Analytics & Diagnostics

Aggregates student assessment performance over time to provide actionable placement readiness insights. The analytics engine computes overall average score, category-wise accuracy percentages, and a diagnostic breakdown identifying strong topics (accuracy >= 75%) and weak topics (accuracy < 50%).

4.10 In-Portal Notification & Alert Management

Dispatches in-portal notifications displayed through the notification center for application-status changes and newly published aptitude tests. Students receive notifications when their application status changes (e.g., Shortlisted, Rejected, Selected) or when new aptitude tests are published. Companies receive in-portal notifications upon new applicant submissions.

4.11 Administrator Governance & Portal Monitoring

Provides authorized college placement administrators with master oversight capabilities. Admins review and approve/reject pending company registrations, activate/deactivate user accounts, moderate internship listings, manage question banks, and inspect aggregate portal analytics.

Functional Requirements Specification

Table 4.1 specifies exactly 24 functional requirements for the portal. Each requirement uses mandatory 'shall' phrasing, is assigned a priority (High, Medium, Low), linked to its primary stakeholder, associated with rigorous acceptance criteria, and cross-referenced with a specific verification test case.

Req ID	Functional Requirement Statement (The system shall...)	Type	Priority	Stakeholder	Acceptance Criteria & Test Case Ref	Dependencies & Comments
OIP-F-001	The system shall allow a Student to register an account using a unique institutional email, secure password, full name, and university roll number.	Functional	High	Student	AC-OIP-F-001: Student record created in database with role 'Student' and active status. Test: TC-AUTH-01	Validates email format and unique roll number.

Req ID	Functional Requirement Statement (The system shall...)	Type	Priority	Stakeholder	Acceptance Criteria & Test Case Ref	Dependencies & Comments
OIP-F-002	The system shall allow a Company recruiter to register an account with corporate email, organization name, website domain, and recruiter contact details.	Functional	High	Company / Admin	AC-OIP-F-002: Company record created with 'Pending Approval' status; listing creation disabled until verified. Test: TC-AUTH-02	Requires Admin approval before posting internships.
OIP-F-003	The system shall authenticate registered users (Student, Company, Admin) against encrypted credentials and issue a role-scoped JSON Web Token (JWT).	Functional	High	All Actors	AC-OIP-F-003: Valid credentials return HTTP 200 and signed JWT; invalid credentials return HTTP 401. Test: TC-AUTH-03	Depends on bcrypt password hashing; token valid for 4 hours.
OIP-F-004	The system shall allow an authenticated user to log out securely by invalidating the active session token on the client.	Functional	Medium	All Actors	AC-OIP-F-004: JWT removed from client storage; subsequent protected API calls return HTTP 401. Test: TC-AUTH-04	Prerequisite: Authenticated session.
OIP-F-005	The system shall allow a Student to create, update, and view their academic profile, including department, graduation year, and CGPA bounded between 0.00 and 10.00.	Functional	High	Student	AC-OIP-F-005: Valid academic data persists to MongoDB; CGPA < 0.00 or > 10.00 rejected with HTTP 400. Test: TC-PROF-01	Prerequisite: Authenticated Student session.
OIP-F-006	The system shall allow a Student to add, modify, and delete technical skills in their profile as categorized keyword tags.	Functional	High	Student	AC-OIP-F-006: Skills stored as an array of normalized strings associated with the student profile. Test: TC-PROF-02	Used during automated eligibility matching.
OIP-F-007	The system shall allow a Student to upload and replace a resume file strictly in PDF format with file size not exceeding 5 MB.	Functional	High	Student	AC-OIP-F-007: Non-PDF or >5MB files rejected with HTTP 400; valid PDF stored securely and linked to profile. Test: TC-PROF-03	Uses Multer middleware with MIME validation.
OIP-F-008	The system shall allow a Company to manage its organizational profile, including corporate description, logo URL, headquarters location, and industry domain.	Functional	High	Company	AC-OIP-F-008: Profile updates persist and render on public internship listing cards. Test: TC-COMP-01	Prerequisite: Authenticated Company session.

Req ID	Functional Requirement Statement (The system shall...)	Type	Priority	Stakeholder	Acceptance Criteria & Test Case Ref	Dependencies & Comments
OIP-F-009	The system shall allow an approved Company to create and post an Internship listing specifying title, description, skills, minimum CGPA, stipend, and deadline.	Functional	High	Company	AC-OIP-F-009: Listing saved with status 'Active'; visible to eligible students in catalog. Test: TC-INT-01	Requires Company approval status == 'Approved'.
OIP-F-010	The system shall allow a Company to edit details of an active Internship listing or deactivate/close the listing to stop accepting applications.	Functional	Medium	Company	AC-OIP-F-010: Deactivated listing prevents new submissions; existing applications remain viewable. Test: TC-INT-02	Prerequisite: Listing owned by authenticated Company.
OIP-F-011	The system shall allow a Student to browse active, approved Internship listings in a paginated catalog displaying title, company name, location, and stipend.	Functional	High	Student	AC-OIP-F-011: Displays paginated active listings sorted chronologically by default. Test: TC-INT-03	Only approved and active listings are returned.
OIP-F-012	The system shall allow a Student to search and filter Internship listings by keyword, required technical skill, location, work arrangement, and minimum stipend.	Functional	High	Student	AC-OIP-F-012: Query returns matching listings within 1.5 seconds; empty search returns all active roles. Test: TC-SRCH-01	MongoDB indexes on title, skills, and location.
OIP-F-013	The system shall automatically check a Student's eligibility (CGPA, required skills) before permitting submission of an Internship application.	Functional	High	Student	AC-OIP-F-013: Students failing CGPA or skill criteria receive an explicit eligibility warning. Test: TC-APP-01	Compares student profile against internship criteria.
OIP-F-014	The system shall allow an eligible Student to apply for an Internship and strictly prevent duplicate applications by the same student for the same listing.	Functional	High	Student	AC-OIP-F-014: First application created with status 'Applied'; subsequent attempt rejected with HTTP 400. Test: TC-APP-02	Compound unique index on (student_id, internship_id).
OIP-F-015	The system shall allow a Student to track all submitted applications and view current status (Applied, Under Review, Shortlisted, Rejected, Selected).	Functional	High	Student	AC-OIP-F-015: Dashboard renders chronological table of submitted applications with status badges. Test: TC-APP-03	Prerequisite: Authenticated Student session.

Req ID	Functional Requirement Statement (The system shall...)	Type	Priority	Stakeholder	Acceptance Criteria & Test Case Ref	Dependencies & Comments
OIP-F-016	The system shall allow a Company to view all submitted applications for its posted Internships and securely view applicant profiles and access authorized PDF resumes.	Functional	High	Company	AC-OIP-F-016: Recruiter accesses applicant dossier and accesses authorized PDF resume via authorized endpoint. Test: TC-APP-04	Restricted to listings owned by the company.
OIP-F-017	The system shall allow a Company to update an applicant's status to 'Shortlisted', 'Rejected', or 'Selected', with an optional decision note.	Functional	High	Company	AC-OIP-F-017: Application status updated; triggers an in-portal notification for the applicant. Test: TC-APP-05	Triggers In-Portal Notification module.
OIP-F-018	The system shall generate and deliver in-portal notifications to Students for application status updates and to Companies for new application submissions.	Functional	Medium	Student / Company	AC-OIP-F-018: In-portal notification bell counter increments; clicking marks notification as read. Test: TC-NOTIF-01	Internal notification collection in MongoDB.
OIP-F-019	The system shall allow an Admin to manage the aptitude question bank (create, edit, delete questions across Quant, Logical, and Verbal categories with solutions).	Functional	High	Admin	AC-OIP-F-019: Question records persist with stem, options, correct answer index, and explanation. Test: TC-TEST-01	Restricted strictly to role == 'Admin'.
OIP-F-020	The system shall allow an Admin to create, configure, publish, and deactivate Aptitude Tests specifying title, category, duration, and associated questions.	Functional	High	Admin	AC-OIP-F-020: Configured test saved; publishing test makes it immediately visible to students. Test: TC-TEST-02	Requires active questions in selected category.
OIP-F-021	The system shall enforce a synchronized countdown timer during an Aptitude Test and automatically submit test answers when time expires.	Functional	High	Student	AC-OIP-F-021: Client input locks on timer expiry; test payload automatically submitted to server. Test: TC-TEST-03	Client timer synchronized with server start timestamp.
OIP-F-022	The system shall automatically calculate scores upon test submission, display instant results with answer explanations, and record the attempt in Test History.	Functional	High	Student	AC-OIP-F-022: Score calculated immediately; results view renders total score and correct answers. Test: TC-TEST-04	Persists attempt record to TestHistory collection.

Req ID	Functional Requirement Statement (The system shall...)	Type	Priority	Stakeholder	Acceptance Criteria & Test Case Ref	Dependencies & Comments
OIP-F-023	The system shall compute and display Student Preparation Analytics, showing test history, category-wise accuracy, and identifying strong (>=75%) and weak (<50%) topics.	Functional	Medium	Student	AC-OIP-F-023: Analytics view charts past test scores and renders topic diagnostic breakdown. Test: TC-AN-01	Requires at least one completed test attempt.
OIP-F-024	The system shall allow an Admin to view portal analytics, approve or reject pending Company accounts, and activate or deactivate user accounts.	Functional	High	Admin	AC-OIP-F-024: Admin dashboard displays aggregate counts and allows toggling company approval status. Test: TC-ADMIN-01	Restricted strictly to role == 'Admin'.

5. Non-Functional Requirements (Detailed)

Non-Functional Requirements (NFRs) specify the quality attributes, operational limits, performance targets, and architectural standards governing the portal. Table 5.1 details eight realistic, measurable NFRs covering Performance, Availability/Reliability, Usability, Scalability, Maintainability, Accessibility, Data Integrity, and Compatibility/Portability.

Req ID	Non-Functional Requirement Statement	Category	Priority	Acceptance Criteria & Measurement
OIP-NF-001	The system API shall respond to 95% of standard HTTP read/write requests within 1.5 seconds, and full dashboard pages shall render within 2.5 seconds under standard load.	Performance	High	95th percentile API response time <= 1.5s; page load <= 2.5s under standard simulated testing. Test: TC-PERF-01
OIP-NF-002	The system shall maintain an operational availability of at least 99.5% during the academic evaluation period, with automated daily database backups.	Availability / Reliability	High	System logs demonstrate >= 99.5% uptime; daily MongoDB mongodump archives verified. Test: TC-REL-01
OIP-NF-003	The user interface shall provide intuitive navigation and clear validation messages, allowing a first-time student to apply for an internship in under 3 minutes.	Usability	High	User testing cohort of 10 students completes registration and application submission within 3 minutes. Test: TC-UX-01
OIP-NF-004	The system shall support at least 500 registered student accounts, 50 company accounts, and up to 100 concurrent aptitude test takers without throughput degradation.	Scalability	High	Concurrent load simulation of 100 test takers shows error rate < 1% and average response < 2.0s. Test: TC-PERF-02
OIP-NF-005	The codebase shall maintain modular component decoupling across React frontend and Express backend, adhering to standard REST conventions.	Maintainability	Medium	Modular folder structure (controllers, routes, models); clean separation of concerns confirmed by code review. Test: Code Audit
OIP-NF-006	The user interface shall comply with the Web Content Accessibility Guidelines (WCAG) 2.1 Level AA, including minimum 4.5:1 color contrast and full keyboard navigability.	Accessibility	Medium	Automated axe-core accessibility audit reports zero critical/serious violations. Test: TC-UX-02
OIP-NF-007	The database schema shall enforce relational and referential integrity, strictly preventing duplicate applications and orphaned candidate records.	Data Integrity	High	Compound unique index on (student_id, internship_id) and Mongoose schema constraints verified. Test: TC-APP-02
OIP-NF-008	The web application shall be fully functional across modern versions of Google Chrome, Mozilla Firefox, Apple Safari, and Microsoft Edge on desktop and mobile devices.	Compatibility / Portability	High	Cross-browser testing verifies consistent layout and functionality across Chrome, Firefox, Safari, and Edge. Test: TC-UX-03

5.1 Security Requirements & Objectives

5.1.1 Security Objectives

The portal is designed to fulfill four primary security objectives:

- **1.** SO-01: Protection of Personal and Student Information: Maintain strict confidentiality of student academic transcripts, CGPA, contact details, and uploaded resumes to prevent unauthorized access or data leakage.
- **2.** SO-02: Prevention of Unauthorized Access & Role Enforcement: Ensure that all requests are cryptographically authenticated and restricted strictly to the user's authorized role (Student, Company, Admin) via the principle of least privilege.
- **3.** SO-03: Integrity of Application, Internship, and Test Data: Ensure that test questions, exam scores, internship listings, and candidate application statuses are tamper-proof and modifiable only by authorized entities.
- **4.** SO-04: Non-Repudiation & Operational Traceability: Maintain timestamped audit records for critical events such as company approvals, listing deactivations, and application status transitions.

5.1.2 Security Requirements Specification

Table 5.2 defines six enforceable security requirements governing cryptography, transport security, role-based authorization, input sanitization, session lifetime, and document access.

Req ID	Security Requirement Statement (shall...)	Type	Priority	Acceptance Criteria & Test Case Ref
OIP-SR-001	The system shall hash all user passwords using the bcrypt cryptographic algorithm with a work factor (salt rounds) of at least 10 prior to database storage.	Security	High	AC-OIP-SR-001: Zero plaintext passwords stored in MongoDB; all passwords verified as bcrypt hashes. Test: TC-SEC-01
OIP-SR-002	The system shall enforce Role-Based Access Control (RBAC) middleware on every protected API endpoint, validating user role before granting access.	Security	High	AC-OIP-SR-002: Requests with unauthorized roles (e.g., student accessing admin route) return HTTP 403 Forbidden. Test: TC-SEC-02
OIP-SR-003	The system shall issue cryptographically signed JWT tokens with an expiration lifetime not exceeding 4 hours, rejecting expired or tampered tokens.	Security	High	AC-OIP-SR-003: Expired tokens return HTTP 401 Unauthorized; token tampering detected and rejected. Test: TC-SEC-03
OIP-SR-004	The system shall enforce Transport Layer Security (TLS 1.2 or higher) for all client-server communications when deployed in production.	Security	High	AC-OIP-SR-004: All HTTP traffic redirected to HTTPS; unencrypted endpoints blocked. Test: TC-SEC-04
OIP-SR-005	The system shall validate, sanitize, and escape all incoming client request payloads to prevent Cross-Site Scripting (XSS) and NoSQL injection attacks.	Security	High	AC-OIP-SR-005: Express-validator / sanitizer strips dangerous injection payloads; malicious queries rejected. Test: TC-SEC-05
OIP-SR-006	The system shall restrict uploaded PDF resume access strictly to the owning Student, the applied Company, and the System Administrator.	Security	High	AC-OIP-SR-006: Direct URL access attempts to candidate resumes by unauthorized third parties return HTTP 403 Forbidden. Test: TC-SEC-06

6. Quality Attributes & Acceptance Tests

6.1 Quality Attributes Specification

The portal is designed around five core software quality attributes adapted for a college software engineering project:

- **1.** Performance & Responsiveness: Fast page loads (< 2.5s) and responsive search execution (< 1.5s) using indexed MongoDB queries.
- **2.** Security & Data Protection: Role-based access control and company-level authorization, bcrypt password hashing, role-based route middleware, and authorized PDF resume access.
- **3.** Reliability & Data Integrity: Robust schema validation, error handling, unique indexing to prevent duplicate applications, and consistent state transitions.
- **4.** Usability & Accessibility: Mobile-responsive fluid layout, clear status badges, intuitive test navigation, and WCAG 2.1 AA accessibility compliance.
- **5.** Maintainability & Clean Architecture: Modular MVC/REST folder organization with clear separation between routes, controllers, models, and UI components.

6.2 Acceptance and Exit Criteria

The criteria for formal acceptance and deployment readiness of the portal are defined as follows:
1. Full Functional Coverage: 100% of all 24 Functional Requirements (OIP-F-001 through OIP-F-024) implemented and verified.
2. Security Verification: All 6 Security Requirements verified, with zero plaintext passwords and zero unauthorized route bypasses.
3. Usability Verification: First-time student user testing confirms application completion time under 3 minutes.
4. Quality Assurance: Zero open Severity-1 (critical crash/data loss) or Severity-2 (major workflow blockage) defects.
5. Complete Traceability: 100% of requirements mapped to design modules and test cases in the Requirements Traceability Matrix (RTM).

6.3 Acceptance Test Suite Matrix

Table 6.1 specifies the acceptance test cases designed to validate system features, non-functional requirements, and security controls. In accordance with the SRS stage, all test cases are defined and pending execution.

Test Case ID	Target Feature / Module	Test Scope & Objective	Expected Verification Result	Priority
TC-AUTH-01	Student Registration	Verify student registration with valid institutional email and duplicate roll number.	Valid registration creates student account; duplicate roll number returns HTTP 400.	High
TC-AUTH-02	Company Registration	Verify company registration creates account in 'Pending Approval' status.	Company created with Pending status; listing creation blocked until approved.	High
TC-AUTH-03	User Authentication	Verify JWT issuance upon valid credentials and rejection upon invalid password.	Valid credentials return JWT with role claim; invalid credentials return HTTP 401.	High

Test Case ID	Target Feature / Module	Test Scope & Objective	Expected Verification Result	Priority
TC-AUTH-04	User Logout	Verify client token removal and blocking of subsequent protected requests.	Token cleared from client; subsequent API requests return HTTP 401.	Medium
TC-PROF-01	Student Profile	Verify profile updates and validation of CGPA within 0.00–10.00 range.	Valid profile persists; CGPA < 0 or > 10 rejected with HTTP 400.	High
TC-PROF-02	Skills Inventory	Verify adding, editing, and deleting technical skills array.	Skills array persists and updates correctly in MongoDB.	High
TC-PROF-03	Resume Upload	Verify upload of valid PDF <= 5MB and rejection of non-PDF or files > 5MB.	Valid PDF accepted; .exe or >5MB files rejected with HTTP 400.	High
TC-COMP-01	Company Profile	Verify company profile updates and rendering on public listing cards.	Company profile details persist and display correctly.	High
TC-INT-01	Internship Creation	Verify approved company can create listing with title, CGPA, and deadline.	Listing saved as 'Active'; visible in student internship catalog.	High
TC-INT-02	Internship Updates	Verify company can edit listing metadata and deactivate/close listing.	Deactivated listing prevents new submissions; remains viewable to existing applicants.	Medium
TC-INT-03	Internship Catalog	Verify paginated browsing of active approved internship postings.	Returns paginated list of active listings sorted chronologically.	High
TC-SRCH-01	Search & Filter	Verify multi-parameter search by keyword, skill, location, and stipend.	Returns filtered results matching all criteria within 1.5 seconds.	High
TC-APP-01	Eligibility Check	Verify eligibility engine blocks students whose CGPA is below requirement.	Ineligible student receives clear warning; eligible student allowed to apply.	High
TC-APP-02	Application Submit	Verify application binds Student+Internship+Company and prevents duplicates.	First application succeeds (status Applied); second attempt rejected with HTTP 400.	High
TC-APP-03	Application Track	Verify student can track all submitted applications and status changes.	Chronological list of applications rendered with current status badges.	High
TC-APP-04	Applicant Review	Verify company can view applicant dossier and access authorized PDF resume.	Recruiter accesses candidate details and accesses authorized PDF resume via authorized route.	High
TC-APP-05	Status Transition	Verify company can update applicant status to Shortlisted, Rejected, or Selected.	Status updated in DB; in-portal notification dispatched to student.	High
TC-NOTIF-01	In-Portal Notif	Verify in-portal notification delivery upon status change and test publication.	Notification bell badge increments; clicking marks notification as read.	Medium

7. System Models and UML Diagrams

This section models the operational interactions and conceptual relationships of the Online Internship & Placement Preparation Portal. It provides two comprehensive UML Use-Case Diagrams—one for the Student subsystem and one for the Company & Administrator subsystems—alongside complete PlantUML source code, embedded rendered diagram graphics, and a rigorous conceptual domain relationship analysis.

7.1 UML Use-Case Diagram 1: Student Subsystem

The Student subsystem encompasses all self-service workflows available to technical students, ranging from initial registration and resume repository management to multi-parameter internship discovery, application submission, and aptitude test preparation. In strict accordance with clean UML design principles, direct actor-to-use-case associations are used to avoid unnecessary relationship clutter.

PlantUML Source Specification (Student Subsystem)

```
@startuml
left to right direction
skinparam packageStyle rectangle
skinparam monochrome false
skinparam shadowing false

actor Student as "Student"

rectangle "Online Internship & Placement Preparation Portal (Student Subsystem)" {
  usecase "Register" as UC_S1
  usecase "Login / Logout" as UC_S2
  usecase "Manage Profile" as UC_S3
  usecase "Manage Education & Skills" as UC_S4
  usecase "Upload / Manage Resume" as UC_S5
  usecase "Browse Internships" as UC_S6
  usecase "Search / Filter Internships" as UC_S7
  usecase "View Internship Details" as UC_S8
  usecase "Apply for Internship" as UC_S9
  usecase "Track Application Status" as UC_S10
  usecase "Take Aptitude Test" as UC_S11
  usecase "View Test Results" as UC_S12
  usecase "View Test History" as UC_S13
  usecase "View Preparation Analytics" as UC_S14
  usecase "Receive Notifications" as UC_S15
}

Student --> UC_S1
Student --> UC_S2
Student --> UC_S3
Student --> UC_S4
Student --> UC_S5
Student --> UC_S6
Student --> UC_S7
Student --> UC_S8
Student --> UC_S9
Student --> UC_S10
Student --> UC_S11
Student --> UC_S12
```

```
Student --> UC_S13
Student --> UC_S14
Student --> UC_S15
@enduml
```

Embedded Rendered Diagram (Student Subsystem)

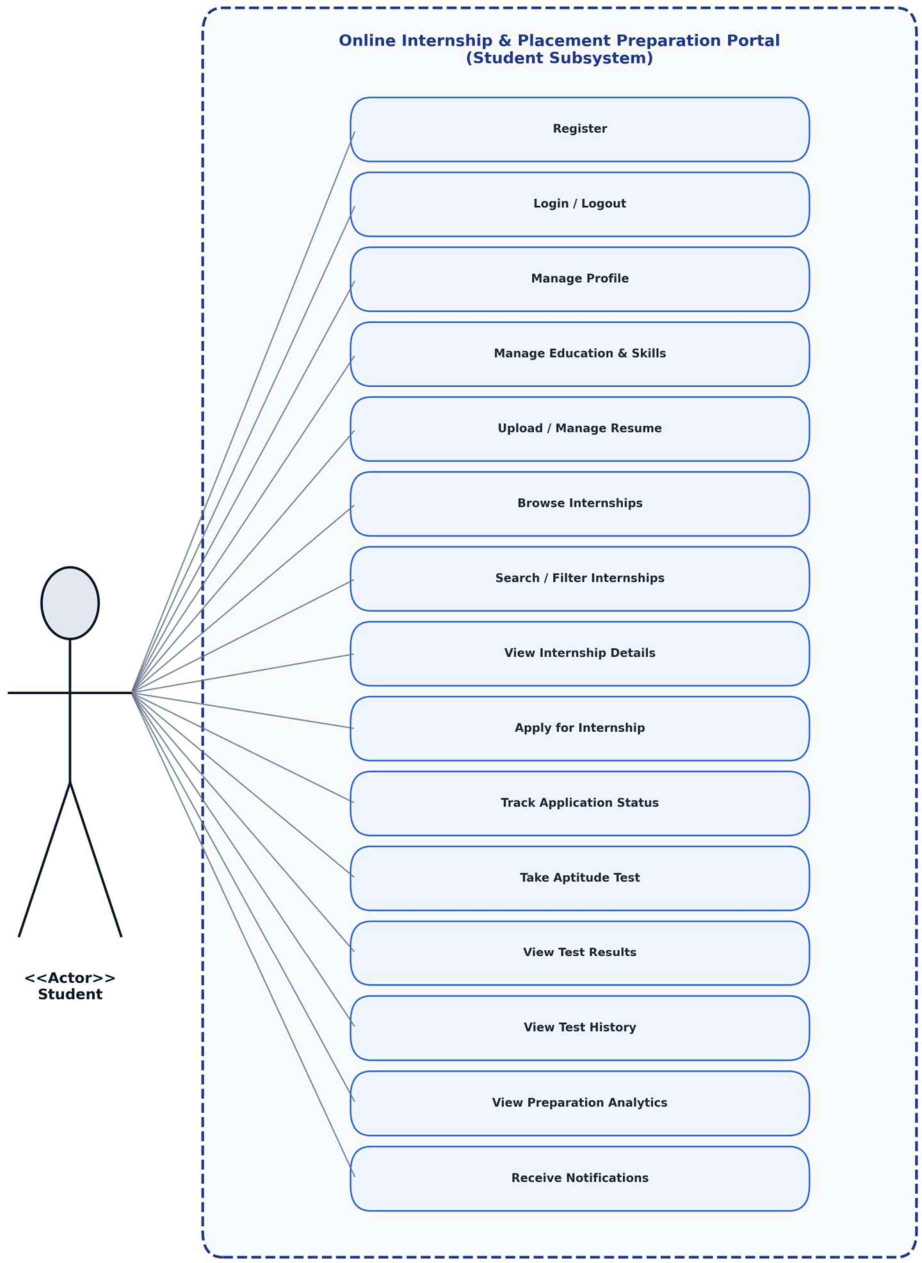

Figure 7.1: UML Use-Case Diagram for the Student Subsystem

7.2 UML Use-Case Diagram 2: Company and Admin Subsystems

The Company and Administrator subsystems model recruitment management and governance workflows. Companies manage internship listings and applicant reviews, while Administrators oversee corporate authenticity, listings moderation, question bank curation, test publishing, and portal monitoring.

PlantUML Source Specification (Company & Admin Subsystem)

```
@startuml
left to right direction
skinparam packageStyle rectangle
skinparam monochrome false
skinparam shadowing false

actor Company as "Company"
actor Admin as "Admin"

rectangle "Online Internship & Placement Preparation Portal (Company & Admin Subsystems)" {
  package "Company Use Cases" {
    usecase "Register / Login" as UC_C1
    usecase "Manage Company Profile" as UC_C2
    usecase "Post Internship" as UC_C3
    usecase "Edit Internship" as UC_C4
    usecase "Deactivate Internship" as UC_C5
    usecase "View Applications" as UC_C6
    usecase "View Applicant" as UC_C7
    usecase "View Resume" as UC_C8
    usecase "Shortlist Applicant" as UC_C9
    usecase "Reject Applicant" as UC_C10
    usecase "Update Application Status" as UC_C11
  }

  package "Admin Use Cases" {
    usecase "Login" as UC_A1
    usecase "Manage Student Accounts" as UC_A2
    usecase "Manage Company Accounts" as UC_A3
    usecase "Approve / Reject Company" as UC_A4
    usecase "Manage Internship Listings" as UC_A5
    usecase "Manage Question Bank" as UC_A6
    usecase "Create / Configure Aptitude Tests" as UC_A7
    usecase "Manage Test Availability" as UC_A8
    usecase "View Portal Analytics" as UC_A9
    usecase "Monitor System Activity" as UC_A10
  }
}

Company --> UC_C1
Company --> UC_C2
Company --> UC_C3
Company --> UC_C4
Company --> UC_C5
Company --> UC_C6
Company --> UC_C7
Company --> UC_C8
Company --> UC_C9
```

```
Company --> UC_C10
Company --> UC_C11

Admin --> UC_A1
Admin --> UC_A2
Admin --> UC_A3
Admin --> UC_A4
Admin --> UC_A5
Admin --> UC_A6
Admin --> UC_A7
Admin --> UC_A8
Admin --> UC_A9
Admin --> UC_A10
@enduml
```

Embedded Rendered Diagram (Company & Admin Subsystem)

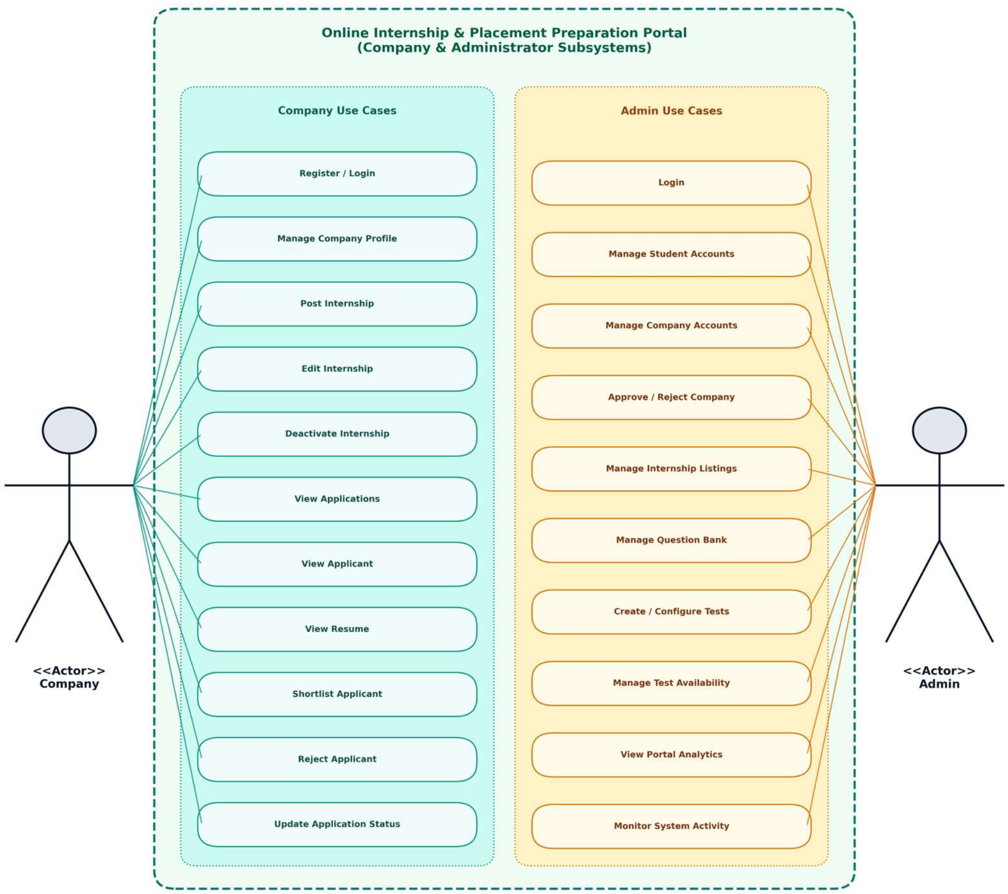

Figure 7.2: UML Use-Case Diagram for Company and Administrator Subsystems

7.3 Conceptual Domain Relationship: Student → Application → Internship → Company

A foundational design principle of the portal is the explicit modeling of the recruitment domain. The central transactional entity is the Application, which acts as the associative bridge binding Students, Internships, and Companies.

CARDINALITY & RELATIONAL BUSINESS RULES

1. Company to Internship (1 : N) — One Company can post many Internships. Each Internship is created and owned strictly by exactly one Company.
2. Student to Application (1 : N) — One Student can submit many Applications to distinct internship opportunities. However, a Student can submit at most ONE Application per unique Internship listing (duplicate prevention).
3. Internship to Application (1 : N) — One Internship listing can receive many candidate Applications over its publication lifecycle.
4. Application Associative Relationship — An Application connects exactly one Student, one Internship, and one Company. It encapsulates candidate snapshot data, resume link, submission timestamp, and current recruitment status (Applied, Under Review, Shortlisted, Rejected, Selected).

Figure 7.3 depicts the conceptual domain data flow:

```
┌──────────────┐   submits (1:N)   ┌──────────────────┐   targets (N:1)   ┌────────────────┐
posted by (N:1)   ┌─────────────┐
|  STUDENT  | ─────────────────────> |  APPLICATION  | ─────────────────────> |  INTERNSHIP
| ─────────────────────────> |  COMPANY  |
|      |           |       |           |       |           |      |
| • Student_ID|          | • App_ID     |          | • Int_ID     |          | • Comp_ID  |
| • Profile  |          | • Student_Ref  |          | • Company_Ref |          | • Org Name |
| • CGPA     |          | • Internship_Ref|          | • Title, CGPA |          | • Sector   |
| • Resume_URL|          | • Status, Date |          | • Skills, Deadl|          | • Status   |
└──────────────┘          └─────────────────┘          └────────────────┘
└──────────────┘
```

8. Requirements Traceability Matrix (RTM)

The Requirements Traceability Matrix (RTM) establishes forward and backward traceability across the engineering lifecycle. It maps each Functional Requirement (OIP-F-001 through OIP-F-024), Non-Functional Requirement (OIP-NF-001 through OIP-NF-008), and Security Requirement (OIP-SR-001 through OIP-SR-006) to its respective SRS section, design module specification, verification test case, and verification status. In strict accordance with the baseline Phase 1 SRS stage, all requirements are categorized as 'N' (Not implemented / not yet verified).

Status Legend: N = Not implemented / not yet verified.

Req ID	Requirement Short Description	Section Ref / Design Spec	Module	Test Case(s)	Status	Comments / Verification
OIP-F-001	Student Registration	4.1 / DS-AUTH-01	AuthModule	TC-AUTH-01	N	Validates institutional email & roll number
OIP-F-002	Company Registration	4.1 / DS-AUTH-02	AuthModule	TC-AUTH-02	N	Initial Pending Approval state
OIP-F-003	Secure Authentication & JWT	4.1 / DS-AUTH-03	AuthModule	TC-AUTH-03	N	Role-scoped token claims
OIP-F-004	User Logout	4.1 / DS-AUTH-04	AuthModule	TC-AUTH-04	N	Token removal from client
OIP-F-005	Student Profile Management	4.2 / DS-PROF-01	ProfileModule	TC-PROF-01	N	CGPA range bounded 0.00–10.00
OIP-F-006	Student Skills Inventory	4.2 / DS-PROF-02	ProfileModule	TC-PROF-02	N	Categorized technical skill tags
OIP-F-007	Resume Upload & PDF Check	4.2 / DS-PROF-03	ResumeModule	TC-PROF-03	N	Strict 5MB PDF validation
OIP-F-008	Company Profile Management	4.3 / DS-COMP-01	CompanyModule	TC-COMP-01	N	Recruiter details & company branding
OIP-F-009	Internship Posting Creation	4.4 / DS-INT-01	InternshipModule	TC-INT-01	N	Requires approved company status
OIP-F-010	Internship Listing Updates	4.4 / DS-INT-02	InternshipModule	TC-INT-02	N	Edit details or deactivate listing
OIP-F-011	Internship Catalog Browsing	4.5 / DS-INT-03	InternshipModule	TC-INT-03	N	Paginated display of active roles
OIP-F-012	Search & Multi-filtering	4.5 / DS-SRCH-01	SearchEngine	TC-SRCH-01	N	Keyword & multi-attribute filter
OIP-F-013	Eligibility Verification	4.6 / DS-APP-01	AppEngine	TC-APP-01	N	Pre-submission CGPA & skill check
OIP-F-014	Application Submission	4.6 / DS-APP-02	AppEngine	TC-APP-02	N	Duplicate prevention enforced
OIP-F-015	Student Application Tracking	4.6 / DS-APP-03	AppEngine	TC-APP-03	N	Real-time status tracking

Req ID	Requirement Short Description	Section Ref / Design Spec	Module	Test Case(s)	Status	Comments / Verification
OIP-F-016	Company Applicant Review	4.6 / DS-APP-04	RecruitModule	TC-APP-04	N	Authorized PDF resume access
OIP-F-017	Shortlisting & Status Update	4.6 / DS-APP-05	RecruitModule	TC-APP-05	N	Triggers in-portal notification
OIP-F-018	In-Portal Notification System	4.10 / DS-NOTIF-01	NotificationModule	TC-NOTIF-01	N	In-portal alert bell & badge count
OIP-F-019	Admin Question Bank Management	4.7 / DS-TEST-01	AptitudeModule	TC-TEST-01	N	Quant, Logical, Verbal with explanations
OIP-F-020	Admin Aptitude Test Config	4.7 / DS-TEST-02	AptitudeModule	TC-TEST-02	N	Test creation & publishing
OIP-F-021	Timed Test Engine & Timer	4.8 / DS-TEST-03	TestEngine	TC-TEST-03	N	Auto-submit on countdown expiry
OIP-F-022	Instant Scoring & Solutions	4.8 / DS-TEST-04	TestEngine	TC-TEST-04	N	Score calculation & test history log
OIP-F-023	Preparation Progress Analytics	4.9 / DS-AN-01	AnalyticsModule	TC-AN-01	N	Strong/weak topic diagnostic breakdown
OIP-F-024	Admin Account Governance	4.11 / DS-ADM-01	AdminModule	TC-ADMIN-01	N	Company approval & portal monitoring
OIP-NF-001	Response Time <= 1.5s	5.1 / DS-PERF-01	WebAPI / DB	TC-PERF-01	N	95th percentile under normal load
OIP-NF-002	Availability 99.5% & Backup	5.2 / DS-REL-01	InfraOps	TC-REL-01	N	Daily MongoDB backups verified
OIP-NF-003	Usability Task Completion < 3m	5.3 / DS-UX-01	Frontend UI	TC-UX-01	N	Student usability evaluation
OIP-NF-004	Scalability 100 Concur. Tests	5.4 / DS-PERF-02	InfraOps / API	TC-PERF-02	N	100 concurrent test submissions
OIP-NF-005	Maintainability Modular MERN	5.5 / DS-MAINT-01	DevOps / CI	Code Audit	N	MVC separation of concerns
OIP-NF-006	Accessibility WCAG 2.1 AA	5.6 / DS-UX-02	Frontend UI	TC-UX-02	N	Color contrast & keyboard tests
OIP-NF-007	Data Integrity & No Duplicates	5.7 / DS-DB-01	DatabaseEngine	TC-APP-02	N	Unique compound index verified
OIP-NF-008	Compatibility Modern Browsers	5.8 / DS-UX-03	Frontend UI	TC-UX-03	N	Chrome, Firefox, Safari, Edge verified
OIP-SR-001	bcrypt Password Hashing	5.1.2 / DS-SEC-01	AuthModule	TC-SEC-01	N	Salt rounds >= 10 verified
OIP-SR-002	RBAC Route Authorization	5.1.2 / DS-SEC-02	Middleware	TC-SEC-02	N	Role verification per endpoint

Req ID	Requirement Short Description	Section Ref / Design Spec	Module	Test Case(s)	Status	Comments / Verification
OIP-SR-003	JWT Session Security	5.1.2 / DS-SEC-03	AuthModule	TC-SEC-03	N	Tokens expire within 4 hours
OIP-SR-004	TLS 1.2+ Transport Security	5.1.2 / DS-SEC-04	NetworkInfra	TC-SEC-04	N	HTTPS enforced in production
OIP-SR-005	Input Validation & Anti-XSS	5.1.2 / DS-SEC-05	Middleware	TC-SEC-05	N	Sanitizes all payload parameters
OIP-SR-006	Secure Document Access	5.1.2 / DS-SEC-06	ResumeModule	TC-SEC-06	N	Strict anti-IDOR authorization

9. Project Development Plan & Phase Transitions

To ensure successful execution throughout the academic semester, Team 4 has structured the project across five distinct engineering phases. As defined in Document Version 1.0, the project is currently situated in Phase 1.

Phase 1 — Requirements Specification (CURRENT PHASE):
Completion of the Software Requirements Specification (SRS) document, establishing 24 Functional Requirements, 8 NFRs, 6 Security Requirements, Acceptance Test definitions, UML Use-Case models, and the Requirements Traceability Matrix (RTM).

Phase 2 — System Architecture & Design (NEXT PHASE):
Development of the System Architecture, MongoDB Entity-Relationship (ER) design, UML Class diagram, Sequence diagrams for core workflows (Application submission, Timed test execution), Activity diagrams, and RESTful API route specifications.

Phase 3 — Implementation & Coding:
Full-stack MERN implementation: React.js frontend dashboards, Node.js + Express.js backend REST services, MongoDB schema enforcement via Mongoose, JWT authentication middleware, Multer resume handling, and in-portal notification mechanics.

Phase 4 — Verification & Testing:
Execution of the Acceptance Test Suite specified in Section 6: automated API endpoint tests using Jest/Supertest, component UI testing, cross-browser compatibility checks, security role validation, and load testing under simulated concurrency.

Phase 5 — Final Documentation & Project Defense:
Compilation of test execution reports, UI screenshot walk-throughs, RTM status updates from 'N' to 'A', final academic report, and preparation for departmental project evaluation.